

Hierarchical Clustering

Albert Y. Kim

- ▶ Unsupervised methods identify patterns in the data without specifying an outcome measure
 - ▶ k -means clustering finds groupings of similar data-points using *prototypes* i.e. cluster centers
 - ▶ k -means is an example of **partitional clustering** as it produces clusters without any overlap. i.e. it partitions the p -dimensional space

Hierarchical Clustering (overview)

- ▶ **Hierarchical clustering** organizes data-points into a tree-like structure known as **dendrogram** that stores a series of nested (overlapping) groupings
- ▶ There are two major types of hierarchical clustering algorithms:
 1. **Agglomerative** - each data-point begins as its own cluster and pairs of clustered are merged using a *linkage criterion*
 2. **Divisive** - all data-points begin in a single cluster that is recursively subdivided until each data-point is its own cluster
- ▶ We'll focus on agglomerative clustering since divisive clustering algorithms aren't currently offered in `sklearn`

Ex: Animals

In data/animals.csv: $n \times p = 20 \text{ animals} \times 6 \text{ features}$. Here are first 4 (see handout):

	warm_blooded	can_fly	vertebrate	endangered	lives_in_groups	has_hair
ant	0	0	0	0	1	0
bee	0	1	0	0	1	1
cat	1	0	1	0	0	1
caterpillar	0	0	0	0	0	1
chimpanzee	1	0	1	1	1	1

Let's compare the following pairs of animals. How dissimilar are they? i.e. How many features do not match?

1. ant and bee?
2. ant and cat?

Ex: Dissimilarity Matrix

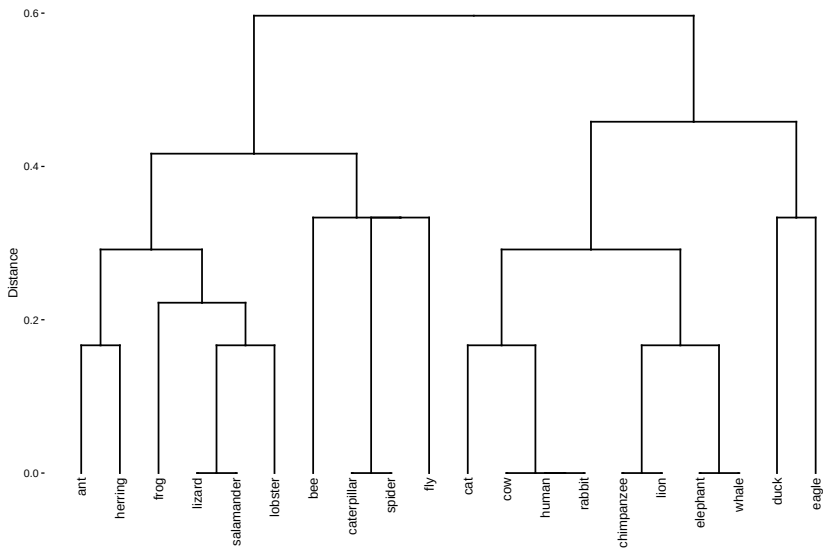
20 × 20 dissimilarity matrix. Here are first 4 × 4 (see handout)

##	ant	bee	cat	caterpillar
## ant	0.000	0.333	0.667	0.333
## bee	0.333	0.000	0.667	0.333
## cat	0.667	0.667	0.000	0.333
## caterpillar	0.333	0.333	0.333	0.000

Ex: Hierarchical clustering represented via Dendrogram

Created via agglomerative clustering using average linkage

Agglomerative clustering of animals (using average linkage)



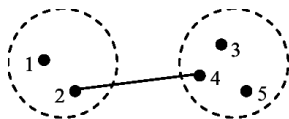
Agglomerative clustering - linkage

Many ways to merge clusters when building tree bottom up

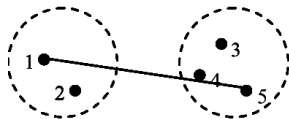
1. *Single linkage*: Find the minimum pairwise distance between data-points in different clusters and merge their clusters
2. *Complete linkage*: Find the maximum pairwise distance between data-points in each pairing of clusters and merge the two clusters with the smallest maximum
3. *Average linkage*: Find the average pairwise distances between points in each pairing of clusters and merge the two clusters with the smallest average distance
4. *Ward's linkage*: Coming up ...

Agglomerative clustering - linkage

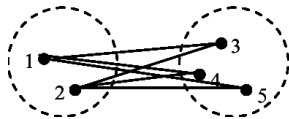
Diagram illustrating single, complete and average linkage. See we have n points already assigned to k clusters. Out of these, let's drill down on 5 points and 2 clusters (dashed circled lines). Each linkage produces a distance



$$d_{24}$$



$$d_{15}$$



$$\frac{d_{13} + d_{14} + d_{15} + d_{23} + d_{24} + d_{25}}{6}$$

Agglomerative clustering (Ward linkage)

The most common linkage criterion used in agglomerative clustering is *Ward's linkage* the increase in sum of squared error accumulated by merging clusters.

Choosing a linkage criterion

1. *Single linkage*: Tends to create large chains of one-at-a-time additions but can be good at identifying irregular patterns
2. *Complete linkage*: Robust to outliers and tends to favor similarly sized clusters at each “level” of the dendrogram
3. *Average linkage*: Lower variability than complete linkage but more impacted by outliers
4. *Ward's linkage*: Clusters tend to be the most compact (desirable) but the method is also the most computationally expensive (a lot of distances need to be calculated)

Agglomerative clustering vs. k -means

- ▶ k -means forms *spherical clusters* (thus it excels with the “blobs” data) but struggles to identify irregularly shaped clusters (ie: “moons” data). Agglomerative clustering tends to be more flexible when applied to unusual data sets
- ▶ Agglomerative clustering can also better handle outliers and do not involve random initialization
- ▶ The main downside of agglomerative clustering is its computational burden when n is large